

Debraj Paul

Technical Lead II — Backend & Platform Engineering

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Dubai, UAE · Indian national · **In Dubai now — UAE Job Seeker Visa, no sponsorship transfer needed**

Open to: Senior / Staff / Principal Engineer · Technical Architect · On-site Dubai · Hybrid · Remote

11+ yrs

Distributed backend systems
Media · fintech · food-tech · travel

~50%

Latency reduction
Swiggy Dineout — POS event pipelines

95%

Credit workflow automation
PT Tujuh Asia — P2P lending platform

80%

Straight-through GDPR/CCPA automation
100% manual → minutes/request, 10+ microservices

PROFESSIONAL SUMMARY

Backend and Platform Technical Lead with 11+ years designing, building, and operating distributed systems across media, fintech, food-tech, and travel. Promoted to Technical Lead II at Warner Bros. Discovery in January 2026. Deep hands-on expertise in Node.js/TypeScript, AWS-native architectures (Lambda, ECS, Glue, OpenSearch), Kafka-based event-driven pipelines, and GraphQL Backend-for-Frontend patterns. Architect of compliance-grade systems: GDPR/CCPA privacy-request orchestration across 10+ microservices — taking fulfilment from 100% manual to ~80% straight-through and per-request turnaround from a day to minutes — plus multi-brand async content moderation and production-aware backfill data pipelines. Strong emphasis on system reliability, API governance, cross-team platform standardisation, and observable, production-ready software. Currently in Dubai on a UAE Job Seeker Visa — available on-site now, no sponsorship transfer required to start.

CORE SKILLS

Languages & Runtime: TypeScript, Node.js (8+ yrs production), JavaScript, Java, Go

Backend & APIs: Express.js, GraphQL (BFF), REST, Spring Boot, Gin — API design and governance at org scale

Cloud (AWS): Lambda, ECS, SQS/SNS (Standard + FIFO + DLQ), S3, DynamoDB, RDS, IAM, Glue, OpenSearch, CloudWatch

Event-Driven & Streaming: Kafka (two-stream contracts, schema evolution), AWS Kinesis, SQS/SNS — at-least-once delivery, idempotency design

Data & Storage: MongoDB / Mongoose, PostgreSQL / MySQL, DynamoDB, Redis, Elasticsearch, OpenSearch, Parquet (Snappy) on S3

Architecture Patterns: Service-registry fan-out · Two-stream Kafka contracts · State-machine async fulfillment · Offline-extract ETL · Cache-aside · Idempotent dedup · Dual-store split (S3 + DynamoDB) · Hot-reloadable per-tenant policy

Infrastructure & CI/CD: AWS CDK, CloudFormation, Docker, Jenkins, GitHub Actions, CircleCI

Compliance & Security: GDPR, CCPA, OneTrust, subject-rights-request orchestration, audit trails, secure data deletion and access controls

Observability: OpenTelemetry, AWS CloudWatch, structured logging, distributed tracing, proactive alerting

Testing: TDD, Jest (9+ yrs), Mocha-Chai, Sinon, Newman, JUnit

PROFESSIONAL EXPERIENCE

Technical Lead II — Backend & Platform Engineering · Warner Bros. Discovery (via Robosoft Technologies)

Aug 2020 – Present · Remote (Global)

► Global content platform · multi-brand (Food Network, CNN, Eurosport, Magnolia, Golf Digest, MotorTrend) · Node.js/TypeScript · Kafka · AWS CDK · GDPR/CCPA-critical

Project: GDPR/CCPA Privacy Request Orchestrator

- Designed and owned a compliance-grade async orchestrator fulfilling GDPR Article 15 (access) and Article 17 (erasure) requests across **10+ PII-holding microservices** — took fulfilment from **100% manual to ~80% straight-through**, cutting per-request turnaround from ~a day to minutes.
- Implemented a service-registry pattern: the orchestrator reads a registry to determine which downstream services hold PII per user, enabling new services to join compliance by registering — with zero orchestrator changes.
- Applied a two-stream Kafka pattern (request-out + result-in topics) with a subject-request-ID as the idempotency key end-to-end — re-deliveries at any hop are safe.
- State machine tracks per-service status and an aggregate; aggregate flips to complete only when every individual settles, gating the fulfilment callback to the privacy broker. Access exports staged in S3; broker receives a signed URL.

Project: Multi-Brand Async Content Moderation Platform

- Migrated text moderation off a third-party vendor (cost + policy ownership) and absorbed image moderation from a sister team — built and owned both in-house classifiers across a brand-agnostic submission contract.

- Designed a two-stream Kafka architecture (Moderation Request / Response) with per-hop replay isolation — a failure in one classifier never re-drives the full pipeline.
- Implemented hot-reloadable per-brand policy (banned terms, severity thresholds) — trust-and-safety teams ship rule changes without redeploying the moderator. Applied dual-store: image bytes in S3, record metadata + S3 keys in DynamoDB.
- Serving UGC (comments, nicknames, image uploads) from N consumer brand surfaces (web + mobile) with CIAM identity verification at the submission boundary.

Platform-Level Contributions

- Defined and standardised REST and GraphQL API design conventions adopted across all engineering teams — reducing cross-team integration friction and onboarding time.
- Implemented distributed tracing (OpenTelemetry) across services, improving incident root-cause analysis. Introduced Redis cache-aside strategy reducing DynamoDB read pressure.
- Architected AWS serverless and container infrastructure (Lambda, ECS) using AWS CDK — standardising multi-environment deployments and enabling fault-isolated, cost-optimised scaling.
- Mentored engineers across squads through code reviews, design sessions, and architectural guidance — raising the collective engineering quality bar.

Senior Software Engineer — Payments & Integrations Platform · [Swiggy Dineout](#)

Dec 2019 – Jun 2021 · Bengaluru, India

► *High-volume food-tech · Kafka in production · 10+ POS integrations · event-driven pipelines · revenue-critical*

- Designed and built a high-performance integration platform connecting **10+ third-party POS systems** using Kafka and Redis — reduced transaction latency by **~50%** through async event pipelines and optimised caching.
- Built customer segmentation logic on MongoDB and Elasticsearch, enabling behavioural targeting (frequency, spend, activity) — contributed to **~25% conversion lift** and **~15% bounce-rate reduction**.
- Implemented marketing trigger systems delivering personalised campaigns on real-time segmentation. Designed Kafka topic schemas and a backwards-compatible event-evolution strategy.
- Delivered three enterprise-grade AWS microservices contributing to **~25% YoY revenue growth**.

Senior Software Engineer — Fintech Platform · [PT Teknologi Tujuh Asia](#)

Nov 2018 – Dec 2019 · Jakarta, Indonesia

► *P2P lending · KYC automation · risk decisioning · Node.js/AWS*

- Automated **~95% of credit-processing workflows** — improved loan-approval accuracy by **~25%** and reduced end-to-end processing time by **~50%** through system-driven decision pipelines.
- Built OCR + facial-recognition pipelines for KYC and identity verification; developed an NLP-based scoring engine for risk analysis — reduced communication errors by **~90%**, improved customer satisfaction by **~20%**.

Software Engineer (Backend) · [TripBorn](#) · [Ekagga Technology](#) · [Freelance](#)

Jan 2015 – Oct 2018 · Bengaluru

- Built booking/reservation backends (TripBorn), enterprise analytics APIs on Java/Spring Boot + PostgreSQL (Ekagga), and end-to-end REST API solutions for multiple clients (Freelance). Full SDLC ownership across all engagements.

SELECTED OPEN-SOURCE PROJECTS

Full Budget App — Multi-tenant fintech SaaS budgeting platform on AWS serverless (pnpm monorepo, Node 24, strict TS, AWS CDK). Event-driven ingestion (S3 → SQS → Lambda; DynamoDB Streams → Lambda), hybrid deterministic-NLP + Amazon Bedrock (LLM) categorisation behind cost-aware feature flags, schema-first GraphQL (Apollo Server 5), JWT with refresh-token rotation + reuse detection + token family revocation, production-grade observability (X-Ray, CloudWatch, structured logging).

github.com/debrajpaul/full-budget-app · MIT

Resort-Booking MCP Connector — Open-source Model Context Protocol connector turning a booking workflow into Claude-callable tools: research → plan → validate → book, with explicit human confirmation gating every side-effectful action. Local-first (Docker, HTTP/SSE MCP server), phased path to AWS/Terraform/EKS. Node.js/TypeScript, MCP TypeScript SDK, Playwright, node-cron. MIT (in progress)

CERTIFICATIONS

- **AWS Certified Developer – Associate (2024–2027)** — Lambda, IAM, SQS/SNS, API integration, security, cloud best practices
- **Apollo GraphQL Developer – Associate (2025)** — schemas, resolvers, query optimisation, BFF patterns

EDUCATION

Master of Computer Applications (MCA) — Tezpur University, Assam, India · 2012 – 2015

Bachelor of Computer Applications (BCA) — Gauhati University, Assam, India · 2009 – 2012